Mohamed Sebbar

Embedded Hardware & Electronics Engineer

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PROFILE

Embedded Hardware and Electronics Engineer with an M.Sc. in Embedded Systems and Computer Engineering, combining system-level architecture with hands-on electronics design, digital hardware, and experimental validation. Experienced in translating engineering requirements into implementable hardware architectures, developing multilayer mixed-signal PCBs, evaluating power and interface constraints, and preparing manufacturing-ready documentation. Work spans STM32-based measurement systems, FPGA platform architecture, RF-enabled embedded electronics, and digital computer architecture. Research on precision current-signature acquisition has resulted in an accepted manuscript in *Engineering Research Express* (IOP Publishing). Particularly interested in technically demanding hardware problems where architectural decisions, circuit implementation, and measurable performance must align.

ENGINEERING EXPERIENCE

Embedded Hardware Engineer – Independent Contract *Oct. 2026 – Present*

- Lead the hardware architecture and requirements definition of an industrial Spartan-7 FPGA-based acquisition platform, translating system-level objectives into a detailed Technical Requirements Document (TRD), interface definitions, and implementation constraints.
- Define the architecture of a 6-layer board integrating dual multichannel 16-bit ADCs, external memory, digital I/O, clocking, reset circuitry, and a 1 Gbps aggregate M-LVDS interface.
- Develop preliminary power and component budgets, evaluate power-distribution solutions using LTspice, and assess grounding, isolation, electrical protection, thermal constraints, and signal-integrity requirements.
- Prepare schematics, BOMs, interface specifications, and fabrication-oriented engineering documentation to support design reviews and subsequent PCB implementation and verification.

Embedded Hardware Engineer – M.Sc. Research Internship *Feb. 2026 – Jul. 2026*

Faculty of Sciences, Ibn Zohr University

- Designed, fabricated, brought up, and experimentally validated a four-layer STM32F411-based platform for precision current-signature acquisition, covering schematic design, component selection, PCB layout, firmware integration, and bench testing.
- Developed the PCB in KiCad, addressing stack-up, grounding, power distribution, return-current paths, and USB Full-Speed differential routing; applied IPC-2221A calculations and controlled-impedance design principles.
- Implemented a measurement chain combining an INA226 monitor, a 0.5 Ω shunt, timestamped acquisition firmware, and USB CDC/DFU connectivity, with filtering, decoupling, and USB ESD protection.
- Produced manufacturing deliverables and conducted board bring-up, functional checks, fault isolation, and experimental characterization, documenting acquisition behavior and measurement limitations.
- Demonstrated timestamped acquisition at approximately 664 Hz in the documented USB setup; the research contributed to an accepted manuscript in *Engineering Research Express*.

SELECTED ENGINEERING PROJECTS

Industrial IoT Monitoring & Automation – Embedded Electronics

- Designed a four-layer ESP32-C3 mixed-signal PCB integrating custom 2.4 GHz RF circuitry, battery charging, power management, environmental sensing, and motor-drive power circuitry.
- Performed component selection, schematic capture, PCB layout, and antenna matching-network design; used LTspice to investigate power-conversion behavior. Hardware assembly and bring-up remain outstanding.

Custom 16-bit Computer Architecture – Digital Design

- Designed and simulated a custom 16-bit computer architecture encompassing CPU logic, control logic, and memory using HDL.
- Developed a two-pass assembler in C and a virtual-machine translator, exploring the relationship between instruction sets, digital logic, and software execution.

CORE TECHNICAL SKILLS

Hardware Architecture & Electronics: System requirements · Component selection · Schematic design · Mixed-signal electronics · Analog measurement circuits · ADC interfaces · Power architecture · DC/DC conversion · Protection and isolation
PCB Design & Implementation: KiCad · Multilayer PCB layout · Stack-up and grounding · Return-current paths · Signal and power integrity fundamentals · Controlled-impedance routing principles · EMC/EMI-aware design · IPC-2221A · BOM · Gerber and drill files · Design-rule checks
Digital Hardware & Embedded Systems: FPGA platform architecture · HDL-based digital design · Computer architecture · STM32 · ESP32 · Bare-metal C · STM32 HAL · USB CDC/DFU · Hardware/software integration · Interface definition
Verification, Analysis & Tools: Hardware bring-up · Functional testing · Experimental validation · Fault isolation · Measurement analysis · LTspice · STM32CubeMX · STM32CubeProgrammer · Oscilloscope · Digital multimeter · Python · MATLAB · Git
Languages: English (C1) · French (B2) · Spanish (A1)

RESEARCH PUBLICATION

Design and Experimental Validation of an STM32-Based Embedded Platform for Non-Invasive Current Signature Acquisition in Anomaly-Detection Applications.
Engineering Research Express, IOP Publishing. Accepted manuscript published online, 20 October 2026.
doi.org/10.1088/2631-8695/aeb6a1

EDUCATION

M.Sc. Embedded Systems & Computer Engineering – Ibn Zohr University, Morocco *2024 – 2026*
B.Sc. Embedded Systems & Computer Engineering – Ibn Zohr University, Morocco *2021 – 2024*